Akoizo Original Practice - Form 5 - Answer key

Division C | Wright Stuff | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) What does a horizontal stabilizer mainly provide?

Unit: Stability, trim, and circling flight.
Worked-case data: A plane stalls on every glide test. List adjustments in order to try.
Separate investigation observations:
Glide: Smooth and straight.
Powered launch: Plane rolls left and spirals down in the first 5 s.
After 5 s: Torque drops; plane recovers if still airborne.

A. Pitch stability

B. Thrust

C. Roll control

D. Lift for the whole plane

Correct: A

2. (2 points) Why do many teams wind to a target torque?

Unit: Rubber motors: storing and releasing energy.
Worked-case data: A team uses a 15:1 winder and tests show a motor breaks near 1,400 turns. How many cranks should they wind for a safe 8 percent back-off?
Separate investigation observations:
Motor A: 1.4 g, 30 cm long; 1,100 turns; climbed to ceiling, 78 s.
Motor B: 1.4 g, 38 cm long; 1,350 turns; stayed below ceiling, 104 s.
Ceiling: 8 m in both tests.

A. Torque predicts the climb rate better than turns alone

B. It is required

C. It breaks fewer winders

D. It adds energy

Correct: A

3. (2 points) Which propeller is NOT allowed?

Unit: Rules, scoring, and the flight log bonus.
Worked-case data: A team with a complete flight log flies official flights of 96.4 s and 112.0 s. What is its score and tiebreaker?
Separate investigation observations:
Team A: Best flight 120 s, no flight log.
Team B: Best flight 105 s, complete flight log.
Team C: Best flight 112 s, partial flight log.

A. One whose blade angle changes by an active mechanism during flight

B. A purchased fixed-pitch propeller

C. A team-built balsa propeller

D. A propeller with a freewheel

Correct: A

4. (2 points) Which practice keeps trim consistent between flights?

Unit: Building light and strong.
Worked-case data: A built airframe weighs 7.4 g. The CG is slightly behind the target. How should the team meet the 8.0 g minimum?
Separate investigation observations:
Flight 1: Fresh build, 102 s.
Flight 6: After a week in a hot car, 61 s; plane turned sharply right.
Inspection: Right wing twisted 2 degrees.

A. Changing propellers often

B. Retensioning film between flights

C. Marking and keying removable wing and tail positions

D. Storing the plane loose in a bag

Correct: C

5. (4 points) Explain why blade angle changes from hub to tip.

Unit: Propellers: turning torque into thrust.
Worked-case data: A plane cruises at 1.2 m/s and its propeller effectively advances 0.24 m per turn. A motor has 1,200 usable turns. Estimate motor run time.
Separate investigation observations:
Prop 1: Diameter 22 cm; plane cruised level; motor ran 140 s.
Prop 2: Diameter 26 cm, same pitch; plane sank after the burst; 90 s.
Prop 3: Diameter 18 cm; plane climbed fast; motor ran 95 s.

answer: The tip moves faster than the hub, so a flatter angle at the tip makes every section advance the same distance per turn. (4 points).

6. (4 points) Why do indoor planes use long, narrow wings?

Unit: Wings and slow flight.
Worked-case data: A 9.5 g plane (with motor) has 0.040 m^2 of wing and C_L = 0.8. Find its level-flight speed (ρ = 1.2 kg/m^3).
Separate investigation observations:
Plane 1: Wing area 0.030 m^2; cruise 2.6 m/s; flight 85 s.
Plane 2: Wing area 0.042 m^2; cruise 2.1 m/s; flight 118 s.
Same: Mass, motor, and turns.

answer: High aspect ratio reduces induced drag, which dominates at the low speeds indoor planes fly. (4 points).

7. (4 points) Why record bad flights in the log?

Unit: Testing, the flight log, and competition day.
Worked-case data: Log data at a 9 m ceiling: torque 0.10 -> 95 s, 0.13 -> 118 s, 0.16 -> 104 s (ceiling bumps). Plan the first official flight.
Separate investigation observations:
Room A, 7 m ceiling: Best time at torque 0.10.
Room B, 11 m ceiling: Best time at torque 0.15.
Competition: Ceiling announced as 9 m.

answer: They show limits and causes of failure, which help choose safe settings and count toward the 10-flight minimum. (4 points).

8. (4 points) Why make only one trim change at a time?

Unit: Stability, trim, and circling flight.
Worked-case data: A plane stalls on every glide test. List adjustments in order to try.
Separate investigation observations:
Glide: Smooth and straight.
Powered launch: Plane rolls left and spirals down in the first 5 s.
After 5 s: Torque drops; plane recovers if still airborne.

answer: Several changes at once hide which one caused the result; single changes build reliable knowledge. (4 points).

9. (4 points) What should the log record for each motor?

Unit: Rubber motors: storing and releasing energy.
Worked-case data: A team uses a 15:1 winder and tests show a motor breaks near 1,400 turns. How many cranks should they wind for a safe 8 percent back-off?
Separate investigation observations:
Motor A: 1.4 g, 30 cm long; 1,100 turns; climbed to ceiling, 78 s.
Motor B: 1.4 g, 38 cm long; 1,350 turns; stayed below ceiling, 104 s.
Ceiling: 8 m in both tests.

answer: Length, width or cross-section, mass, turns, launch torque, and resulting flight time. (4 points).

10. (6 points) Solve this case and explain your reasoning: A team with a complete flight log flies official flights of 96.4 s and 112.0 s. What is its score and tiebreaker?

reasoning: Flight scores: 96.4 x 1.2 = 115.68; 112.0 x 1.2 = 134.40.
Score = best flight score = 134.40.
Tiebreaker = the non-scored official flight score, 115.68. (3 points).

conclusion: Score 134.40 with a 115.68 tiebreaker. (3 points).

11. (6 points) Solve this case and explain your reasoning: A built airframe weighs 7.4 g. The CG is slightly behind the target. How should the team meet the 8.0 g minimum?

reasoning: Mass needed: 8.0 - 7.4 = 0.6 g; add a small margin, e.g., 0.7 g.
Place ballast near the nose to move the CG forward.
Recheck balance and glide after adding it. (3 points).

conclusion: Add about 0.7 g at the nose: it meets the minimum and fixes balance. (3 points).

12. (2 points) Why did Prop 2 give a short flight?

Supplied synthetic observations:
Prop 1: Diameter 22 cm; plane cruised level; motor ran 140 s.
Prop 2: Diameter 26 cm, same pitch; plane sank after the burst; 90 s.
Prop 3: Diameter 18 cm; plane climbed fast; motor ran 95 s.

A. It needed more torque than the cruise portion supplied

B. It spun too fast

C. It was too light

Correct: A

13. (2 points) Define “Lift equation” in the context of Wright Stuff.

answer: $L = \frac{1}{2}\rho v^2 S C_L$. (2 points).

14. (2 points) Define “Controlled variable” in the context of Wright Stuff.

answer: Factor held constant during a test. (2 points).

15. (2 points) Define “Neutral point” in the context of Wright Stuff.

answer: CG position at which pitch stability vanishes. (2 points).

16. (2 points) Define “Torque burst” in the context of Wright Stuff.

answer: High torque right after launch. (2 points).

17. (2 points) Define “Official flight” in the context of Wright Stuff.

answer: Declared scoring flight; up to two per team. (2 points).

18. (2 points) Define “Ballast” in the context of Wright Stuff.

answer: Added mass to adjust balance or meet minimum mass. (2 points).